EFU AGRICULTURE LICENCE v1.0

Experimental EFU-based Agricultural & Sovereignty Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for sustainable agriculture (arable, horticulture, livestock).

NOT a legal production permit, nitrates directive substitute, or organic certification replacement, but an open research protocol measuring agricultural resource use in human-scale EFU metrics (water, fertilizer, soil, energy).

I. Core Concepts

1. **EFU-Agri (EFU-AGRI)**
   EFU framework application for agriculture:
   - **EFU-W_AGRI**: irrigation water + rainfall supplementation
   - **EFU-N_AGRI**: nitrogen fertilizer (N2O emissions equivalent)
   - **EFU-P_AGRI**: phosphorus (depleting resource)
   - **EFU-H_AGRI**: energy (tractors, machinery, dryers)
   - **EFU-SOIL_AGRI**: soil degradation (erosion, compaction)
2. **Agricultural Sovereignty Gap (SS_AGRI)**
   SS_AGRI = TEFU_AGRI_total – Regional Golden Fertility Carrying Capacity (RFCC_AGRI)
   Measures if production stays within soil regeneration + water capacity limits.
3. **MROI_AGRI**
   MROI_AGRI = THI_AGRI / TEFU_AGRI
   THI_AGRI = yield + soil regeneration + food security + biodiversity.

II. EFU Agriculture Licence – Research Status

Documents EFU-audited farms/estates with standardized metrics.
Does NOT replace pesticide permits, irrigation concessions, or CAP subsidy conditions.
Enables grain, vegetable, livestock farm benchmarking.

III. Metrics and Thresholds

1. EFU Components (per tonne yield)

EFU-W_AGRI (Water)
EFU-W_AGRI = irrigation_m3 / 18.25 / yield_tonnes
Example: Wheat = 3,000 m³/ha → 3,000/18.25/7t = 23.4 EFU_W/tonne

EFU-N_AGRI (Nitrogen)
EFU-N_AGRI = N_fertilizer_kg / 50kg/person/year / yield_tonnes
(50kg N/person/year global average)

EFU-H_AGRI (Energy)
EFU-H_AGRI = mechanization_fuel_MWh / 12.88 / yield_tonnes

EFU-SOIL_AGRI (Soil)
EFU-SOIL_AGRI = erosion_tonnes / soil_baseline / yield_tonnes
Proxy: monoculture years × erosion factor

TEFU_AGRI = 1.5×EFU-W + 1.8×EFU-N + 1.0×EFU-P + 0.8×EFU-H + 1.2×EFU-SOIL

2. Regional Golden Fertility Carrying Capacity (RFCC_AGRI)

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RFCC_AGRI = Sustainable_yield_tonnes / Population / 0.5t/person/year baseline
Example: Netherlands = 15M t / 17.8M / 0.5 = 1.69 EFU_AGRI/person/year
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IV. Licence Categories (EFU-AGRI Labels)

Level	SS_AGRI	MROI_AGRI	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-AGRI SOVEREIGN**	Within soil regeneration
2	0 < ≤10k EFU	≥1e-3	**EFU-AGRI BALANCED**	Monitorable
3	10k < ≤50k EFU	≥1e-4	**EFU-AGRI DEFICIT**	Soil depletion
4	>50k EFU	<1e-4	**EFU-AGRI CRITICAL**	Structural degradation

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V. Regional RFCC_AGRI Calibrations

Country/Region	Yield (M t/year)	Population (M)	RFCC_AGRI (EFU/person/year)
NETHERLANDS	15	17.8	1.69
USA	400	340	0.59
CHINA	650	1,410	0.23
INDIA	300	1,430	0.10
EU AVERAGE	280	448	0.31

VI. Example: Dutch Intensive Vegetable Farming

Input (annual farm):

- Yield: 50,000 tonnes vegetables
- Irrigation: 2M m³ water
- N fertilizer: 1,200 tonnes
- Fuel: 2,500 MWh
- Workers: 150 staff
- Region: Netherlands (RFCC_AGRI = 1.69 EFU/person/year)

Calculation:

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EFU-W_AGRI = 2,000,000/18.25/50,000 = 2.19 EFU_W/tonne
EFU-N_AGRI = 1,200/50/50,000 = 0.00048 EFU_N/tonne
EFU-H_AGRI = 2,500/12.88/50,000 = 0.00388 EFU_H/tonne

TEFU_AGRI_total ≈ 125,000 EFU/year
Allocated = 150 × 1.69 = 253 EFU/year
SS_AGRI = 125,000 – 253 = +124,747 EFU/year (Level 4: CRITICAL)

Result: EFU-AGRI CRITICAL
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VII. Implementation Notes

1. **Nitrates Directive complement**: TEFU_AGRI basis for fertilizer taxation
2. **CAP alignment**: EFU-AGRI Sovereignty indicator for eco-scheme scoring
3. **Data sources**: Tier 1 = irrigation logs, fertilizer bills; Tier 2 = yield diaries